First Sessional Examination 2025 23ECS066

B.Tech.(Computer Engineering) V semester

Microprocessor CSC-502

Marks:-*12* **Time:-One Hour**

Attempt all questions. All question carry equal marks.

CO3 BL3 Q1. Explain what does the following code do when executed:- **2x1.5=3 Marks**

(i)

```
MVI  A, 07H
RLC
MOV B, A
RLC
RLC
ADD B
```

(ii)

```
      MVI A, Byte
      MVI B, 64H
      MVI C, C8H
      CMP B
      JC REJECT
      CMP C
      JNC  REJECT
      OUT  PORT1
      HLT
REJECT: SUB A
      OUT PORT1
      HLT
```

CO3 BL3 Q2. Write a well documented program in assembly language of 8085 to sort a given set of integers stored at consecutive memory locations starting with address 2500H in descending order. First location contains number of item and following addresses contain items. **3 Marks**

CO3 BL3 Q3.Write a well documented program in assembly language of 8085 for 16X8 Division. **3 Marks**

CO3 BL3 Q4. Write a program in assembly language of 8085 to generate Modulo-8 counter with a delay of 100 ms between each count assuming a system with 0.5 microsecond clock period. Show all calculations. **3 Marks**